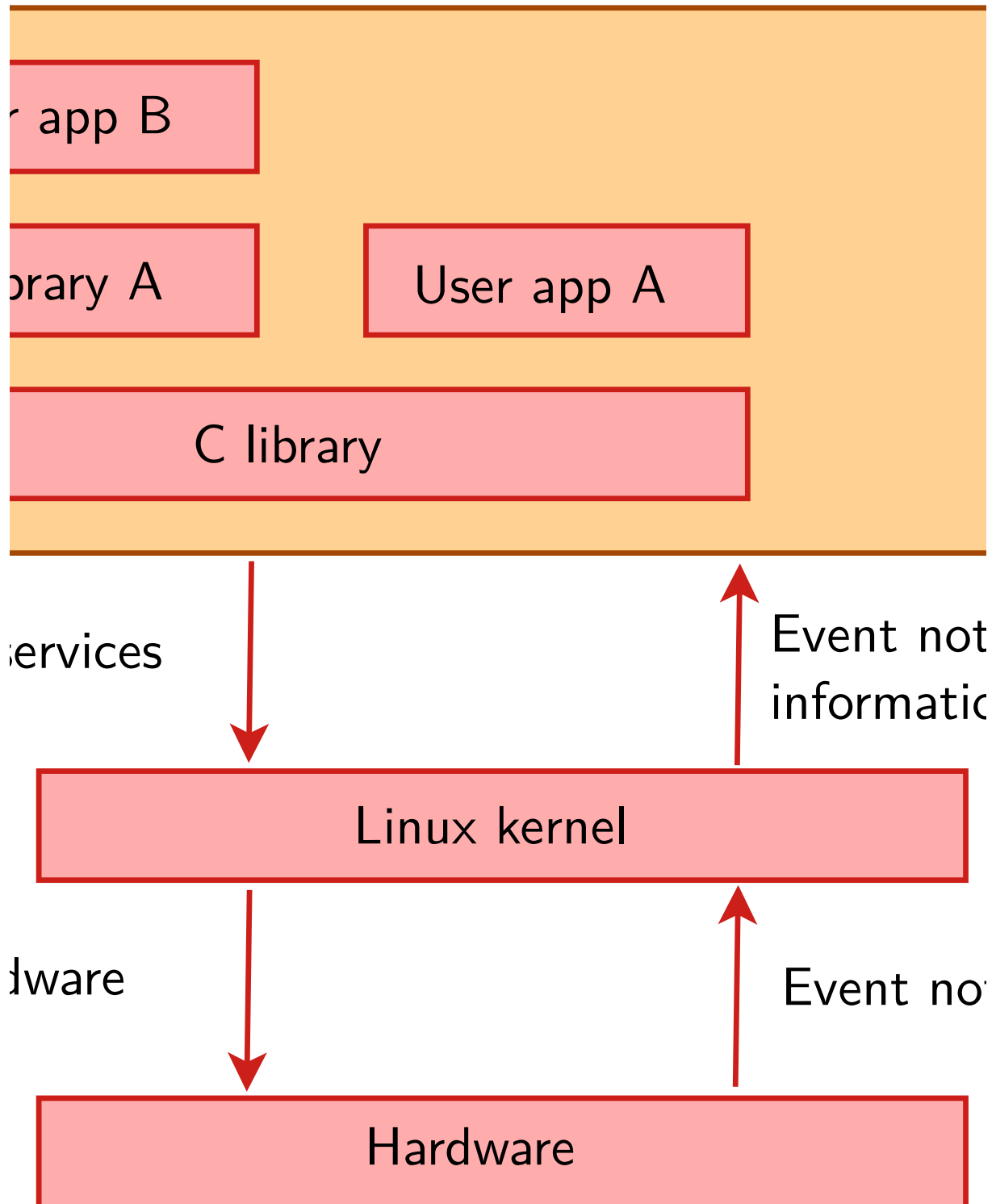


### Linux kernel in the system



### Linux kernel main roles

- **Manage all the hardware resources:** CPU, memory, I/O.
- Provide a **set of portable, architecture and hardware independent APIs** to allow user space applications and libraries to use the hardware resources.
- **Handle concurrent accesses and usage** of hardware resources from different applications.

- Example: a single network interface is used by multiple user space applications through various network connections. The kernel is responsible for “multiplexing” the hardware resource.

### System calls

- The main interface between the kernel and user space is the set of system calls
- About 400 system calls that provide the main kernel services
  - File and device operations, networking operations, inter-process communication, process management, memory mapping, timers, threads, synchronization primitives, etc.
- This system call interface is wrapped by the C library, and user space applications usually never make a system call directly but rather use the corresponding C library function

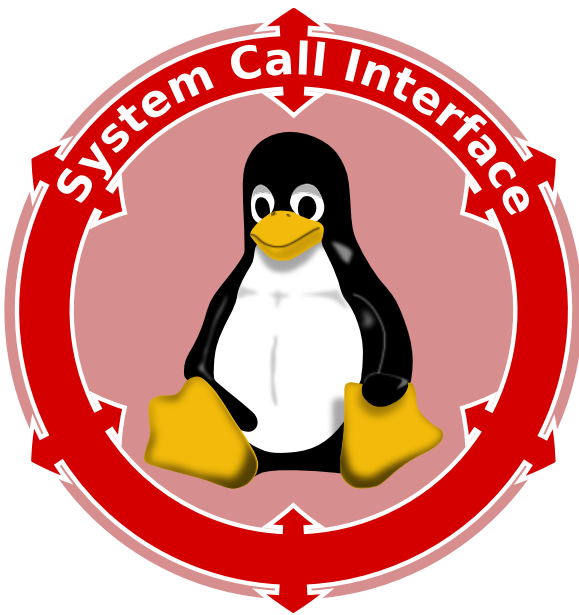


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### Pseudo filesystems

- Linux makes system and kernel information available in user space through **pseudo filesystems**, sometimes also called **virtual filesystems**
- Pseudo filesystems allow applications to see directories and files that do not exist on any real storage: they are created and updated on the fly by the kernel
- The two most important pseudo filesystems are
  - `proc`, usually mounted on `/proc`: Operating system related information (processes, memory management parameters...)

- `sysfs`, usually mounted on `/sys`: Representation of the system as a tree of devices connected by buses. Information gathered by the kernel frameworks managing these devices.